

Hallucination, Calibration, and Missing Mass

Why Error-Free Training Data Do Not Guarantee Error-Free
Answers

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Overview

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- ▶ A generator can be calibrated yet hallucinate. Abstaining avoids false claims but can prevent it from matching the answer-giving source.
- ▶ Missing information quantifies this tension. Facts seen only once help us estimate how much remains unseen.

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The sample can be error-free without being complete.

The Arbitrary-Facts Model Is Deliberately Unlearnable

Let $\mathcal{X}$ be a finite set of prompts with population distribution μ . Fix an integer $K \geq 1$, and give every prompt the same set of factual answers

$$\mathcal{R} = \{1, \dots, K + 1\}.$$

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$(a_x)_{x \in \mathcal{X}}$ independently and uniformly from $\mathcal{R}$.

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$$(a_x)_{x \in \mathcal{X}} \quad \text{independently and uniformly from } \mathcal{R}.$$

Thus learning $a_{x'}$ gives no information about a_x for $x \neq x'$. This models arbitrary facts, not arithmetic, logical consequences, or any fact derivable or predictable from related observations.

Missing Mass Measures How Often the Sample Has No Answer

The training sample contains $n \geq 1$ error-free examples:

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mu, \quad a_{X_1}, \dots, a_{X_n}.$$

Write $O_n = \{X_1, \dots, X_n\}$ for the prompts that appeared.

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$$M_n := \mu(\mathcal{X} \setminus O_n) = \mathbb{P}_{X \sim \mu}(X \notin O_n).$$

It is a population quantity: the probability that a fresh prompt asks for a fact that was absent from training.

Forced Answers Fail on the Missing Mass

Theorem

Suppose an algorithm sees only the training sample, the current test prompt, and its private randomness, and must output one answer in $\mathcal{R}$ on every prompt. Conditional on the realized training prompts,

$$\mathbb{E}[\mathbb{P}_{X \sim \mu}(\hat{a}(X) \neq a_X) \mid X_1, \dots, X_n] \geq \left(1 - \frac{1}{K+1}\right) M_n.$$

The expectation is over the random arbitrary facts and the algorithm's private randomness.

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The expectation is over the random arbitrary facts and the algorithm's private randomness. For birthdays, $K = 364$: on an unseen person, an answer-only system is wrong with probability $364/365$.

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Therefore, whatever distribution the algorithm uses for its answer,

$$\mathbb{P}(\hat{a}(x) = a_x \mid \text{training sample}) = \frac{1}{K + 1}.$$

Its conditional error is $1 - 1/(K + 1)$.

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A fresh prompt is unseen with probability M_n . Keeping only the error on unseen prompts proves the theorem.

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$$\mathcal{Y} := \mathcal{R} \cup \{\perp\}, \quad \perp = \textit{I do not know}.$$

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Missing information does not by itself force
hallucination once abstention is available. We need to
ask what objective the model is trained to satisfy.

Pretraining Introduces a Different Goal

Recall our error-free training source: draw $X \sim \mu$ and report its correct answer a_X . Write D for this same source distribution. With the correct answers fixed,

$$D(r \mid x) = \mathbf{1}\{r = a_x\}, \quad D(\perp \mid x) = 0.$$

Pretraining aims to match D using a generator G learned from the sample and private randomness. On prompt x , it draws $R \sim G(\cdot \mid x)$ on $\mathcal{Y} = \mathcal{R} \cup \{\perp\}$.

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Its hallucination rate counts incorrect factual answers, but not abstention:

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Always abstaining makes $\text{err}_{\text{gen}}(G) = 0$, but mismatches the answer-giving population D . How much distribution matching can we retain without making false claims?

Two Ways to Handle an Unseen Birthday

Return to the birthday example: 365 possible dates, one correct date, and M_n probability of being asked about someone absent from training.

Both generators memorize the birthdays that appeared in the sample.

	unseen person	hallucination rate
Abstain	output $\perp$	0
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In fact, the guessing generator passes a natural calibration test.

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For each prediction value $p \in [0, 1]$, group pairs assigned that value:

$$B_p := \{(x, r) : G(r \mid x) = p\}.$$

Exact *semantic calibration* requires $G(B_p) = D(B_p)$ for every p : each group has the same total mass under the generator and source.

Uniform Guessing Can Be Semantically Calibrated

For the birthday guesser, grouping pairs by $G(r \mid x)$ gives:

prediction value	generator mass	source mass
1: observed correct answers	$1 - M_n$	$1 - M_n$
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Every group matches, yet the hallucination rate is $\frac{364}{365} M_n$.

Matching a group's *total* mass does not locate the correct date inside it.

Abstention Avoids Errors but Breaks Calibration

Now consider the abstaining generator. Recall that O_n is the set of observed prompts. This generator assigns probability 1 to a_x when $x \in O_n$, and to $\perp$ when $x \notin O_n$.

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$$\begin{aligned} B_1 &= \{(x, r) : G(r \mid x) = 1\} \\ &= \underbrace{\{(x, a_x) : x \in O_n\}}_{\text{memorized correct answers}} \cup \underbrace{\{(x, \perp) : x \notin O_n\}}_{\text{certain abstentions}}. \end{aligned}$$

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The generator always outputs a pair in this set, so $G(B_1) = 1$. The source D always supplies a_x . Its output belongs to B_1 on observed prompts, but not on unseen prompts, where B_1 contains $(x, \perp)$ instead. Hence

$$D(B_1) = \mu(O_n) = 1 - M_n.$$

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There are no false claims, but the mass mismatch on B_1 is M_n .

A Weaker Calibration Requirement

Exact calibration matches the source's mass at every prediction value. We will show that matching just one combined group still creates a tradeoff with hallucination. Let H collect all prompt–response pairs assigned probability greater than $1/K$:

$$H := \{(x, r) \in \mathcal{X} \times \mathcal{Y} : G(r \mid x) > 1/K\}, \quad L := H^c.$$

H combines the disjoint groups B_p with $p > 1/K$. Exact calibration therefore implies $G(H) = D(H)$.

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This is also one outcome-indistinguishability test: does a source or generated response fall in H ?

Missing Mass Forces Hallucination or Miscalibration

Theorem

In the arbitrary-facts model, any generator learned from the training sample and private randomness satisfies

$$\mathbb{E}[\text{err}_{\text{gen}}(G) + \delta_H(G, D) \mid X_1, \dots, X_n] \geq M_n - \frac{2}{K}.$$

The expectation is over the arbitrary facts and learner randomness.

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Proof idea: use the generator's probabilities to try to distinguish correct from incorrect answers.

A Validity Test Connects Generation to Missing Information

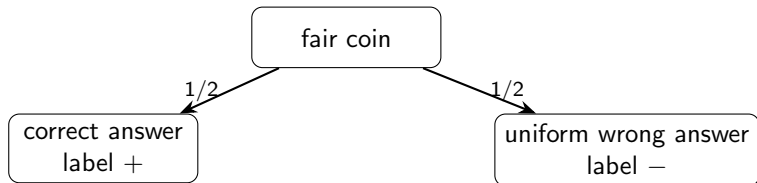
Consider an *is-it-valid* classification experiment:

1. Draw a prompt $X \sim \mu$ and flip a fair coin.
2. On the positive branch, propose $R = a_X$ with label $+$.
3. On the negative branch, propose R uniformly from the K incorrect factual answers, with label $-$.

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This is a *thought experiment*, not extra training data or a procedure the learner must run. It never proposes $\perp$.

On an Unseen Prompt, Both Labels Look the Same

Fix the training sample and an unseen prompt x . Its correct answer a_x is still uniform on $\mathcal{R}$. For any proposed answer r ,

$$\mathbb{P}(R = r \mid +) = \mathbb{P}(a_x = r) = \frac{1}{K+1}.$$

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On the negative branch, r can be proposed for any correct answer $a \neq r$:

$$\mathbb{P}(R = r \mid -) = \sum_{a \neq r} \mathbb{P}(a_x = a) \frac{1}{K} = K \cdot \frac{1}{K+1} \cdot \frac{1}{K} = \frac{1}{K+1}.$$

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The two labels give *identical distributions* of proposed answers. The label remains a fair coin after seeing the proposal, so any classifier using only the training sample and proposal has error $1/2$.

Use Generation Probabilities to Predict Validity

We now use the generator G to define a classifier for the validity-testing problem. Given a prompt x and a proposed factual answer r , predict “valid” if $G(r \mid x) > 1/K$, and “invalid” otherwise. Thus the classifier accepts pairs in H and rejects pairs in L .

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Let $\text{err}_{\text{iiv}}(G)$ be this classifier’s error in the balanced experiment. The unseen-prompt lower bound now applies to this rule:

$$\mathbb{E}[\text{err}_{\text{iiv}}(G) \mid X_1, \dots, X_n] \geq \frac{1}{2} M_n.$$

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To prove the theorem, it therefore suffices to show, for each fixed D, G ,

$$\text{err}_{\text{gen}}(G) + \delta_H(G, D) \geq 2 \text{err}_{\text{iiv}}(G) - \frac{2}{K}.$$

Accepted falsehoods give a direct bound on generated falsehoods; rejected truths give a bound through calibration.

Where Can the Generator Put Its Probability?

We compare classifier errors with generated falsehoods using this partition of prompt–response pairs:

	H : probability $> 1/K$	L : probability $\leq 1/K$
correct	accepted truth	rejected truth
incorrect	accepted falsehood	rejected falsehood
abstain	not validity-tested	not validity-tested

Let $\mathcal{E}$ be the shaded row of incorrect factual pairs. The generator's error is $\text{err}_{\text{gen}}(G) = G(\mathcal{E})$.

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The classifier errs by accepting a falsehood (a *false positive*) or rejecting a truth (a *false negative*). Their probabilities $p_{\text{FP}}, p_{\text{FN}}$ include the fair coin's choice of branch, and

$$\text{err}_{\text{iiv}}(G) = p_{\text{FP}} + p_{\text{FN}}.$$

Accepted Falsehoods Already Have Generated Mass

First consider false positives: incorrect factual pairs in $H \cap \mathcal{E}$. For one such pair (x, r) , its probability in the validity test is

$$\underbrace{\frac{1}{2}}_{\text{negative branch}} \cdot \underbrace{\mu(x)}_{\text{prompt}} \cdot \underbrace{\frac{1}{K}}_{\text{one of } K \text{ wrong answers}} = \frac{\mu(x)}{2K}.$$

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Because of the threshold $1/K$: each accepted falsehood has at least twice as much probability under generation as under the validity test. Summing over these pairs gives

$$\boxed{G(H \cap \mathcal{E}) \geq 2p_{\text{FP}}.}$$

Rejected Truths Require Generated Mass in L

Now consider false negatives: correct answers in L . The positive branch has weight $1/2$, and D generates exactly the correct answers, so

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Because $L = H^c$ and G, D are probability distributions,

$$|G(L) - D(L)| = |G(H) - D(H)| = \delta_H(G, D).$$

Taking the lower bound and then adding $D(L)$ gives

$$\begin{aligned} G(L) - D(L) &\geq -\delta_H(G, D), \\ G(L) &\geq D(L) - \delta_H(G, D) = 2p_{\text{FN}} - \delta_H(G, D). \end{aligned}$$

Rejected Truths Require Generated Mass in L

Now consider false negatives: correct answers in L . The positive branch has weight $1/2$, and D generates exactly the correct answers, so

$$p_{\text{FN}} = \frac{1}{2}D(L).$$

Because $L = H^c$ and G, D are probability distributions,

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Thus a high false negative rate requires high generated probability in L , unless calibration fails.

Separating Incorrect Answers in L

We have shown that

$$G(L) \geq 2p_{\text{FN}} - \delta_H(G, D).$$

$G(L)$ is the probability of generating a pair in L , whether its response is incorrect, correct, or an abstention. To obtain the probability of generating an incorrect answer in L , subtract the probability of generating a correct answer or abstention in L .

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$G(L)$ is the probability of generating a pair in L , whether its response is incorrect, correct, or an abstention. To obtain the probability of generating an incorrect answer in L , subtract the probability of generating a correct answer or abstention in L . Correct answers and abstentions form $\mathcal{E}^c$, so

$$G(L \cap \mathcal{E}) = G(L) - G(L \cap \mathcal{E}^c).$$

We already have a lower bound on $G(L)$. We now need an upper bound on $G(L \cap \mathcal{E}^c)$, the probability being subtracted.

Only Two Outputs per Prompt Avoid Hallucination

For a fixed prompt x , the only pairs in $\mathcal{E}^c$ are (x, a_x) and $(x, \perp)$. If either pair lies in L , its conditional probability under $G(\cdot \mid x)$ is at most $1/K$.

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Combining this upper bound with $G(L) \geq 2p_{\text{FN}} - \delta_H(G, D)$ gives

$$\begin{aligned} G(L \cap \mathcal{E}) &= G(L) - G(L \cap \mathcal{E}^c) \\ &\geq 2p_{\text{FN}} - \delta_H(G, D) - \frac{2}{K}. \end{aligned}$$

Adding the Two Parts Proves the Theorem

Recall $\text{err}_{\text{iiv}}(G) = p_{\text{FP}} + p_{\text{FN}}$. Split the incorrect pairs between H and L , and apply the two bounds:

$$\begin{aligned}\text{err}_{\text{gen}}(G) &= G(H \cap \mathcal{E}) + G(L \cap \mathcal{E}) \\ &\geq 2p_{\text{FP}} + 2p_{\text{FN}} - \delta_H(G, D) - \frac{2}{K} \\ &= 2\text{err}_{\text{iiv}}(G) - \delta_H(G, D) - \frac{2}{K}.\end{aligned}$$

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Now keep the training prompts fixed and average over the random facts and learner randomness. The expected classifier error is at least $M_n/2$, so

$$\mathbb{E}[\text{err}_{\text{gen}}(G) + \delta_H(G, D) \mid X_1, \dots, X_n] \geq M_n - \frac{2}{K}. \quad \square$$

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With large missing mass, matching the source's probability of H forces a large expected hallucination rate.

Why Likelihood Fitting Pushes Toward Calibration

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For each prompt, write

$$h_x := \sum_{r:(x,r) \in H} G(r \mid x)$$

for the probability of a response whose pair lies in H . Multiply the probabilities for pairs in H by $s > 0$, leaving the others unchanged. These are now unnormalized weights: those in H sum to sh_x , and those outside H sum to $1 - h_x$. The total weight over *all responses to prompt* x is therefore $sh_x + (1 - h_x) = 1 + (s - 1)h_x$.

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$$G_s(r \mid x) := \frac{s^{1_{\{(x,r) \in H\}}} G(r \mid x)}{1 + (s - 1)h_x}.$$

At $s = 1$ this is G ; increasing s favors responses in H .

A Probability Mismatch Gives a Loss-Reducing Update

Measure fit by expected negative log-likelihood on source text:

$$\mathcal{L}(s) := \mathbb{E}_{(X,R) \sim D}[-\log G_s(R \mid X)].$$

Assume $\mathcal{L}(1) < \infty$. Taking negative logs of the formula for G_s and averaging gives

$$\mathcal{L}(s) = \mathcal{L}(1) - D(H) \log s + \mathbb{E}_{X \sim \mu}[\log(1 + (s - 1)h_X)].$$

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If $G(H) > D(H)$, slightly decreasing s lowers the loss; if $G(H) < D(H)$, slightly increasing s lowers it. Thus if neither change improves the loss, $G(H) = D(H)$.

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be its sample count. A prompt with $N_x = 1$ is a *singleton*. The Good–Turing estimate is the fraction of the n sample positions occupied by singleton prompts:

$$\hat{M}_n := \frac{1}{n} \sum_{x \in \mathcal{X}} \mathbf{1}\{N_x = 1\}.$$

Each singleton prompt contributes one position; repeated prompts contribute none. Why should this estimate the chance that the next prompt is unseen?

A Singleton Is New When We Hold It Out

The prompt X_j is a singleton exactly when it is absent from the other $n - 1$ positions. Counting singletons by position gives

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$$\begin{aligned} \mathbb{E}[\hat{M}_n] &= \frac{1}{n} \sum_{j=1}^n \mathbb{P}(X_j \notin \{X_i : i \neq j\}) \\ &= \frac{1}{n} \sum_{j=1}^n \mathbb{E}[M_{n-1}] = \mathbb{E}[M_{n-1}]. \end{aligned}$$

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One more example can reduce missing mass. How much?

The Estimate Is High by at Most $1/n$ on Average

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- ▶ Removing a singleton x makes it unseen, increasing missing mass by $\mu(x)$.

Each singleton occupies one of the n positions. The expected increase, conditional on the full sample, is therefore

$$\frac{1}{n} \sum_{x: N_x=1} \mu(x) \leq \frac{1}{n},$$

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since the probabilities of any subset of prompts sum to at most one. The remaining prompts have the distribution of $n - 1$ independent draws. Averaging over the original sample gives

$$0 \leq \mathbb{E}[\hat{M}_n] - \mathbb{E}[M_n] = \mathbb{E}[M_{n-1}] - \mathbb{E}[M_n] \leq \frac{1}{n}.$$

Singletons Quantify the Hallucination Tradeoff

The estimate's average overstatement is at most $1/n$, so $\mathbb{E}[M_n] \geq \mathbb{E}[\hat{M}_n] - 1/n$. Substitute this into the hallucination bound:

$$\begin{aligned}\mathbb{E}[\text{err}_{\text{gen}}(G) + \delta_H(G, D)] &\geq \mathbb{E}[M_n] - \frac{2}{K} \\ &\geq \mathbb{E}[\hat{M}_n] - \frac{1}{n} - \frac{2}{K}.\end{aligned}$$

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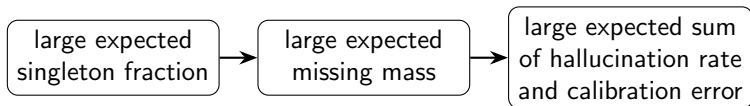
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- ▶ Retrieval or tools can supply information absent from training.
- ▶ Abstaining on unseen facts avoids guessing, but no longer matches the answer-giving source.

Summary

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- ▶ Error-free examples do not supply every fact. An uninformed system can avoid false claims by abstaining.
- ▶ Matching an answer-giving source can conflict with that choice: even a semantically calibrated generator can hallucinate.
- ▶ The chance of encountering an unseen fact controls this tradeoff. Facts observed only once help estimate that chance.

Thanks!

See you next class!

References

- ▶ Surbhi Goel. *Topic 1B: Hallucination, Calibration, and Missing Mass*. CIS 6200: Emerging Theory of LLMs, lecture notes, Fall 2026.
- ▶ Adam Tauman Kalai and Santosh S. Vempala. *Calibrated Language Models Must Hallucinate*. Proceedings of the 56th Annual ACM Symposium on Theory of Computing, 2024.
- ▶ Adam Tauman Kalai, Ofir Nachum, Santosh S. Vempala, and Edwin Zhang. *Why Language Models Hallucinate*. arXiv:2509.04664, 2025.
- ▶ I. J. Good. *The Population Frequencies of Species and the Estimation of Population Parameters*. Biometrika, 1953.
- ▶ David McAllester and Luis Ortiz. *Concentration Inequalities for the Missing Mass and for Histogram Rule Error*. Journal of Machine Learning Research, 2003.